Related Work draft

August 2, 2026

1 Related Work

Chaos forecasting is a challenging field with multiple branches, each with its own models and data analysis. Consider the electrocardiogram (ECG), which has progressed a great deal since the early work on nonlinear cardiac prediction [1], through the breakthrough brought by recurrent architectures such as the LSTM [2], to the deep convolutional and residual models that define current practice [3, 4]. In parallel came the emergence of reservoir computing [5], which is used to enhance prediction at a much lower energy cost: next-generation variants train in milliseconds from only a few hundred samples [6], and in cardiac work specifically, echo state networks and their physics-informed hybrids forecast 15–20 action potentials with action-potential-duration error below 5 ms [7, 8]. Most recently, transformer-based models have been applied to the ECG [9, 10].

We investigated how transformer models have been used for the ECG. The work we found applies them to classification, principally hybrid convolutional–transformer classifiers for inter-patient arrhythmia recognition [9, 10]. So far, however, we have found no work that uses a transformer to forecast the future waveform, which leaves the field open to a novel contribution.

According to the literature the results are still limited, particularly over long horizons. One reason lies in the nature of the data itself, which is chaotic, and which therefore limits forecasting intrinsically.

We therefore turned to another domain, weather, and investigated it. Weather prediction rests on physical models, namely numerical weather prediction systems, of which general circulation models (GCMs) are the climate-scale counterpart. These models which integrate the governing equations directly but require enormous computational power [11]. This motivated the field to ask whether machine learning could deliver comparable predictions at some trade-off, whether in exact pointwise values or in long-term behaviour. That is what we see in NeuralGCM [11], which forecasts for roughly one week at a quality comparable to the physical model, but far faster and at far lower energy cost. Three-dimensional transformer models have also been used [12], but so far the prediction horizon remains on the order of one week.

[Remaining sections to be written.]

References

[1] A. Babloyantz and A. Destexhe. Is the normal heart a periodic oscillator? *Biological Cybernetics*, 58(3):203–211, 1988.

[2] Ö. Yıldırım. A novel wavelet sequence based on deep bidirectional LSTM network model for ECG signal classification. *Computers in Biology and Medicine*, 96:189–202, 2018.

[3] A. Y. Hannun, P. Rajpurkar, M. Haghpanahi, et al. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nature Medicine*, 25:65–69, 2019.

[4] A. H. Ribeiro, M. H. Ribeiro, G. M. M. Paixão, et al. Automatic diagnosis of the 12-lead ECG using a deep neural network. *Nature Communications*, 11:1760, 2020.

[5] H. Jaeger and H. Haas. Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication. *Science*, 304(5667):78–80, 2004.

[6] D. J. Gauthier, E. Bollt, A. Griffith, and W. A. S. Barbosa. Next generation reservoir computing. *Nature Communications*, 12:5564, 2021.

[7] S. Shahi, C. D. Marcotte, C. J. Herndon, F. H. Fenton, Y. Shiferaw, and E. M. Cherry. Long-time prediction of arrhythmic cardiac action potentials using recurrent neural networks and reservoir computing. *Frontiers in Physiology*, 12:734178, 2021.

[8] S. Shahi, F. H. Fenton, and E. M. Cherry. Prediction of chaotic time series using recurrent neural networks and reservoir computing techniques: A comparative study. *Machine Learning with Applications*, 8:100300, 2022.

[9] Y. Xia, Y. Xiong, and K. Wang. A transformer model blended with CNN and denoising autoencoder for inter-patient ECG arrhythmia classification. *Biomedical Signal Processing and Control*, 86:105271, 2023.

[10] Y. Zhou and X. Wang. Heartbeat classification by combining multi-branch convolutional neural networks and transformer. *iScience*, 27(3):109307, 2024.

[11] D. Kochkov, J. Yuval, I. Langmore, et al. Neural general circulation models for weather and climate. *Nature*, 632(8027):1060–1066, 2024.

[12] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619:533–538, 2023.